Results

We recorded ear signals from twenty volunteers during a ninety minute simulated drive. Each volunteer wore a soft earpiece with four dry electrodes and a reference behind the ear. The drive was monotonous on purpose, with long straight roads and very little traffic. Drowsiness was rated by two observers who watched the face camera without seeing the signals.

Signal quality was good for most of the session and degraded mainly when volunteers touched the earpiece. We removed segments with large movement artefacts before any further analysis. On average this removed six percent of each recording.

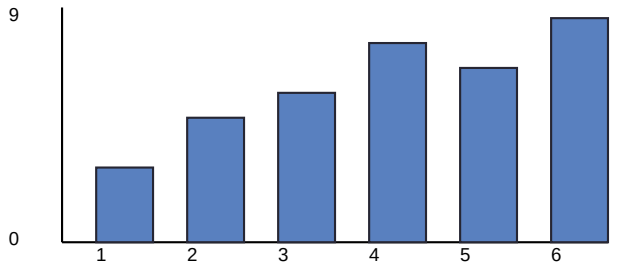

Fig. 4: Drowsiness ratings over time. Ratings rise steadily after the first half hour of driving.

The observers agreed on the drowsiness rating in most of the one minute windows. Where they disagreed, the difference was almost always a single step on the rating scale. We used the mean of the two ratings as the training target for the detector.

Table 2 | Volunteer summary.

Group	Count	Age	Hours
Novice drivers	8	22	12
Regular drivers	9	34	13.5
Professional	3	45	4.5

Novice drivers became drowsy earlier than regular drivers in almost every session. Professional drivers stayed alert the longest, although the group was small. These differences are consistent with earlier studies of simulated driving.

Table 1 | Relevant system and recorder specifications.

Specification	Value
Maximum recording channels	64
Recording channels used	11
ADC resolution	15 bits
Wireless data rate	2 Mbps
Battery life	44 h

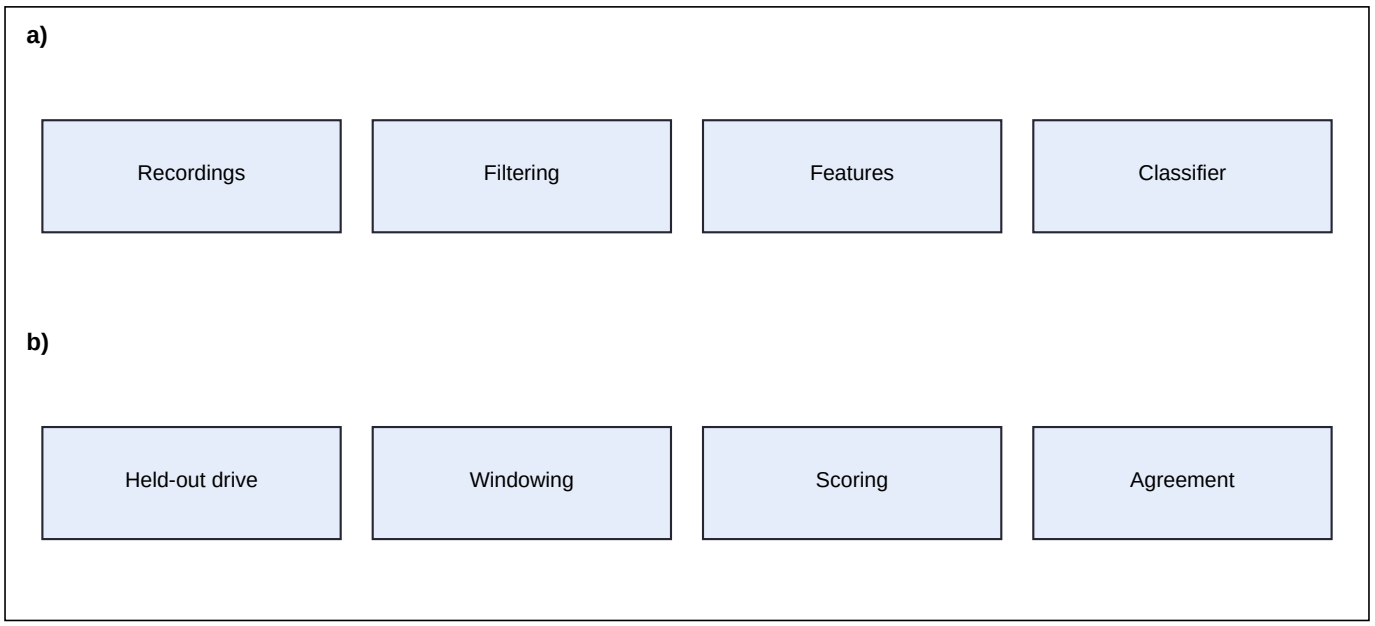

Fig. 5: Drowsiness detector training and validation diagram. The upper panel shows training and the lower panel shows validation on held-out drives.

The detector was trained on one minute windows from sixteen volunteers. Each window was filtered, cut into short epochs and summarised by a small set of spectral features. A gradient boosted classifier then predicted the mean observer rating for the window.

Validation used the four remaining volunteers, whose drives were never seen during training. The predicted ratings agreed with the observers about as well as the observers agreed with each other. The largest errors occurred in the first minutes of each drive, when ratings were low and noisy.